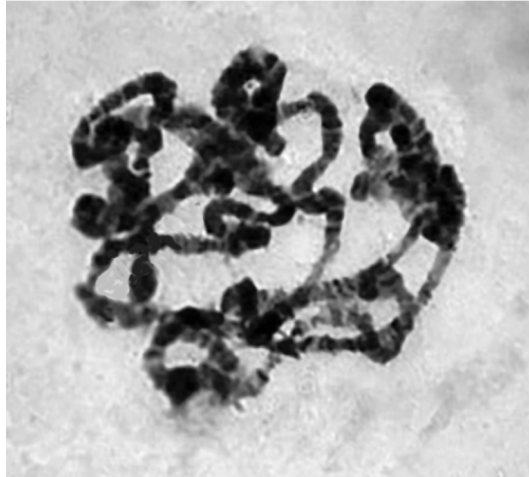


0	4
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This question is about mitosis in cells.

Figure 4 shows the arrangement of the genetic material in a cell during prophase.

Figure 4



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Describe and explain the arrangement of the genetic material shown in **Figure 4**.

[2 marks]

0 4 . 2 The diploid number of chromosomes in the body cell of an insect species is four.

Tick (✓) the box next to the diagram **A**, **B**, **C** or **D** that represents the appearance of chromosomes in a cell during metaphase in this species.

[1 mark]

A		☐
B		☐
C		☐
D		☐

0 4 . 3 Name the fixed position occupied by a gene on a DNA molecule.

[1 mark]

0 4 . 4 Describe how a gene is a code for the production of a polypeptide. Do **not** include information about transcription or translation in your answer.

[3 marks]

